Decision Support System for Sri Lankan Paddy Cultivation to Optimize Financial Efficiency by Predicting Disease Stages and Aligning Cultivation & Harvest According to Future Supply & Demand

Project ID: R26-IT-066

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan paddy farmers suffer recurring yield losses from Rice Blast and Brown Spot. These two most prevalent fungal diseases in Sri Lankan paddy fields yet no existing system quantifies their financial impact on harvest yield in real time. Farmers currently cannot determine whether the cost of treating a detected disease is justified by the yield it would recover, leading to uninformed decisions that worsen income instability.

This research proposes a **Disease-Aware Real-Time Paddy Harvest Yield Estimation** system that predicts expected harvest yield under three scenarios: healthy crop, Rice Blastaffected and untreated, and Brown Spot-affected and treated. The system integrates realtime IoT sensor data from the field monitoring module (IT22178718) and disease severity scores from the disease detection module (IT22261632) with IRRI-validated disease-yield loss models to generate dynamic scenario-based forecasts.

No existing Sri Lankan or global platform integrates real-time IoT field data with IRRIvalidated disease-yield loss models to produce scenario-based harvest yield estimates specifically for Rice Blast and Brown Spot under local paddy conditions. This constitutes the core research gap addressed by this work.

Core technologies include Python with Random Forest and XGBoost for base yield estimation, polynomial regression models trained on IRRI Standard Evaluation System data

for disease-yield loss calculation, FastAPI for REST-based cross-module integration, and a React.js dashboard for scenario visualization. The system is calibrated for Sri Lankan paddy varieties BG 300 and AT 362 across Maha and Yala seasons.

Expected outcomes are a base yield prediction accuracy of 85% or above and disease-yield loss model R-squared of 0.80 or above. The system benefits over 1.8 million Sri Lankan paddy farming households by enabling evidence-based treatment decisions. This work contributes to SDG 2 (Zero Hunger) and SDG 8 (Decent Work and Economic Growth).

Keywords: *paddy harvest yield estimation, rice blast yield loss, brown spot yield loss, IoT-integrated forecasting, disease-aware prediction, Sri Lankan paddy, scenario-based decision support*

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I extend my appreciation to my group members, IT22178718 (IoT Field Monitoring), IT21811104 (Crop Recommendation and Market Analysis), and IT22261632 (Disease Detection and Severity Analysis) whose modules provide the real-time data inputs that this component depends on. The disease severity outputs from IT22261632 in particular form the primary driver of the scenario differentiation that is the core contribution of this research.

I acknowledge the International Rice Research Institute (IRRI) for their publicly available Standard Evaluation System publications, and the Department of Agriculture Sri Lanka and Department of Census and Statistics for historical paddy yield data used in model development and validation.

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LIST OF ABBREVIATIONS

DSS - Decision Support System
IoT - Internet of Things
ML - Machine Learning
RF - Random Forest
XGB - XGBoost - Extreme Gradient Boosting
IRRI - International Rice Research Institute
SES - Standard Evaluation System
DOA - Department of Agriculture Sri Lanka
DCS - Department of Census and Statistics
API - Application Programming Interface
REST - Representational State Transfer
SDG - Sustainable Development Goal
WBS - Work Breakdown Structure
SLIIT - Sri Lanka Institute of Information Technology

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1. INTRODUCTION

1.1 Background and Context

Sri Lankan paddy agriculture is the primary livelihood of over 1.8 million farming households and forms the backbone of national food security [1]. Paddy is cultivated across two seasons annually Maha (October - March) and Yala (April - September) covering approximately 800,000 hectares each Maha season and 479,000 hectares each Yala season [2]. Despite this scale, small and medium-scale farmers consistently face income instability driven by unpredictable yield losses.

Among the most destructive causes of yield loss are Rice Blast (Magnaporthe oryzae) and Brown Spot (Bipolaris oryzae) the two most prevalent fungal diseases in Sri Lankan paddy cultivation. Rice Blast causes yield losses ranging from 10% to 30% under moderate infection, with potential losses reaching 100% in severe outbreaks within 15-20 days of initial infection [3]. Brown Spot causes yield losses of 20% - 40% depending on disease severity and the growth stage at which infection occurs [4]. In the Maha 2023/2024 season alone, an estimated 77,489 hectares of paddy were damaged by disease, pests, and floods representing over 10% of total cultivated area [2].

The farmers most severely affected are smallholders with limited financial reserves who cannot afford to make incorrect treatment decisions. When a disease is detected, farmers face a binary choice: treat at a cost of LKR 1,000-2,500 per acre or delay, risking accelerating yield loss. Without a quantified estimate of the yield recovery that treatment would produce, this decision is made on experience or intuition alone.

Current smart agriculture solutions in Sri Lanka including eKeth [5], GeoGoviya [6], and Krushi Advisor [7] provide static cultivation calendars, generic agro-climate advisories, and offline fertilizer recommendations. None of these platforms integrate real-time IoT field data with disease severity information to produce quantified harvest yield estimates under disease-affected scenarios. The disease detection module in this group project (IT22261632) detects Rice Blast and Brown Spot and estimates severity, but does not translate these outputs into financial harvest impact leaving the economic dimension of disease management unaddressed.

This research proposes a Disease-Aware Real-Time Paddy Harvest Yield Estimation system. The fourth component of the group project R26-IT-066 that acts as the economic intelligence layer of the integrated Smart Agriculture DSS. By combining real-time IoT sensor data, disease severity scores, and IRRI-validated yield loss models, it generates three-scenario harvest yield forecasts that enable farmers to make quantified, evidencebased treatment decisions.

This work directly supports SDG 2 (Zero Hunger) by reducing preventable crop losses, SDG 8 (Decent Work and Economic Growth) by improving farmer income predictability, and SDG 12 (Responsible Consumption and Production) by enabling rational allocation of treatment resources. It aligns with the SST - Software Systems and Technologies research cluster.

1.2 Literature Review

Research directly relevant to this component spans four areas: paddy disease detection and severity assessment, yield prediction modelling, disease-yield loss reference data, and existing Sri Lankan agricultural platforms.

On disease detection, Jayasinghe and Ranasinghe [1] achieved 89.3% classification accuracy using Sri Lankan field imagery, while Wang et al. [2] demonstrated a hybrid IoT-ML detection approach with high accuracy. However, both studies terminate at disease classification without severity grading or yield loss quantification. Chen et al. [3] and Mandal et al. [4] confirmed that disease severity percentage at specific growth stages correlates directly with yield loss independently validating the severity-to-yield regression approach adopted in SO2 of this component.

On yield prediction, Abdurrahman and Miyim [5] validated the Random Forest methodology at 85%+ accuracy, and Ekanayake et al. [6] confirmed environmental parameters as reliable yield predictors across seven Sri Lankan paddy regions. Both studies, however, exclude disease severity entirely as a predictor variable. Peng et al. [7] and DCS Sri Lanka [8] further confirm that Rice Blast reduces yield by 10–30%+ and that over 77,489 hectares were damaged in Maha 2023/2024 establishing the scale of the problem and providing the primary training dataset for SO1.

The IRRI Standard Evaluation System [9] provides validated severity-to-yield-loss tables for Rice Blast and Brown Spot stratified by growth stage, forming the core training data for SO2. Gumma et al. [10] further justify the inclusion of growth stage at detection as a stratification variable, confirming that it significantly moderates yield loss severity.

Existing Sri Lankan platforms e-Krushi [11] and DOA crop guidelines [12] provide static agronomic reference information without real-time detection, severity assessment, or economic decision support. Across all reviewed sources, no system integrates real-time IoT data with disease severity scores to generate scenario-based harvest yield estimates the gap this component addresses.

Author(s)	Institution / Source	Contribution	Key Limitation
Jayasinghe & Ranasinghe (2021	Sri Lankan farms, North Western Province / Ceylon Journal of Science	89.3% classification accuracy on locally collected Sri Lankan paddy field images	Binary healthy/diseased only; no severity grading or yield loss quantification
Wang et al.	International	Hybrid IoT-ML detection approach with high accuracy	Terminates at disease classification; no severity or yield impact output

Ekanayake et al. (2021)	Wayamba University of Sri Lanka	Strong weather-yield relationships validated across 7 Sri Lankan paddy regions	Disease severity not included as a predictor variable
Abdurrahman & Miyim	International	Validated Random Forest & XGBoost for paddy yield prediction (85%+ accuracy)	Excludes disease severity entirely; historical data only
IRRI Standard Evaluation System (SES)	International Rice Research Institute	Validated severity-to-yield loss reference tables stratified by growth stage	Not integrated with real-time sensor data or calibrated for Sri Lankan varieties (BG 300 & AT 362)
Department of Census & Statistics (DCS) & DOA Platforms (e-Krushi, GeoGoviya, Krushi Advisor)	Government of Sri Lanka / Department of Agriculture	National paddy yield statistics and static agro-advisory platforms	Aggregate/static data only; no real-time IoT + disease severity + scenario-based harvest yield & revenue forecasting
Chen et al. & Mandal et al.	International	Confirmed disease severity percentage at specific growth stages correlates directly with yield loss	No real-time IoT integration or financial (revenue) impact calculation

1.3 Research Gap

The literature review reveals a critical and consistent gap across all reviewed sources: no existing system in Sri Lanka or globally combines real-time IoT field data with validated disease-severity-to-yield-loss models to generate scenario-based harvest yield estimates for Rice Blast and Brown Spot under local paddy conditions. This gap can be decomposed into four specific dimensions, each evidence-based:

#	**Identified Gap**	**Evidence from Literature**	**How This Component Address It**
G1	Disease detection systems classify and grade diseases but do not translate detection outputs into quantified harvest yield loss figures.	Wang et al. [8], Chen et al. [10] both achieve high detection accuracy with no yield impact output.	Develops disease-yield loss regression models that map IT22261632 severity scores to yield loss % per growth stage.
G2	Yield prediction models use historical weather data without integrating real-time disease severity or live IoT sensor inputs.	Abdurrahman & Miyim [9]RF model on historical climate; no disease input. DCS[2] aggregate static only.	Combines live IoT sensor data (IT22178718) with disease severity scores (IT22261632) as coinputs to the yield model.
G3	No Sri Lankan research has developed a disease-scenario yield estimation model calibrated for Rice Blast and Brown Spot affecting local varieties BG 300 and AT 362 under Maha/Yala conditions.	eKeth [5], GeoGoviya [6], Krushi Advisor [7] none address disease-yield quantification. DCS [2] provides yield data but not disease-disaggregated.	Calibrates disease-yield loss models on IRRI SES data fine-tuned with Sri Lankan district yield records and local variety susceptibility profiles.
G4	No existing platform enables farmers to compare expected yield and revenue between treating and not treating a detected disease, preventing evidence-based treatment investment decisions.	All reviewed systems. None provide treatment vs. non-treatment scenario comparison with financial output.	Generates three parallel scenarios (Healthy, Untreated, Treated) with revenue projections and net treatment benefit calculation.

Table 02: Research gap Justification

These four gaps together justify the novelty of this component. The research contribution is not incremental improvement to any existing system. It is the first integration of these dimensions into a unified, Sri Lanka-calibrated, economically actionable harvest yield estimation module.

2. OBJECTIVES

2.1 Main Objective

To develop a disease-aware, IoT-integrated paddy harvest yield estimation system that predicts expected harvest yield in kilograms per acre under healthy, Rice Blast-affected, and Brown Spot-affected conditions achieving a base yield prediction accuracy of 85% or above and a disease-yield loss model R-squared of 0.80 or above by integrating real-time IoT sensor data from the field monitoring module (IT22178718) and disease severity scores from the disease detection module (IT22261632), and delivering three-scenario yield and revenue forecasts through a web-based dashboard to enable evidence-based treatment decisions for Sri Lankan smallholder paddy farmers.

2.2 Specific Objectives

The following specific objectives are measurable, aligned with the four identified research gaps (G1-G4), and directly map to the methodology phases and system design components:

SO	Gap	Specific Objectives	Measurable Success Criteria
SO1	G2	To develop a machine learning base yield estimation model using Random Forest and XGBoost that predicts paddy harvest yield in kg/acre using real-time IoT sensor inputs (soil moisture, temperature, humidity, NPK, pH, water level) and crop growth stage indicators received from the field monitoring module (IT22178718).	R-squared >= 0.85 on held-out Sri Lanka DOA district yield validation set for BG 300 and AT 362 varieties.
SO2	G1,G3	To develop separate disease-yield loss regression models for Rice Blast and Brown Spot using IRRI SES severityyield loss data stratified by crop growth stage at detection, fine-tuned for Sri Lankan paddy varieties BG 300 and AT 362 and local pathotype characteristics.	R-squared >= 0.80 on IRRI SES benchmark validation; disease-stage interaction term statistically significant ($p < 0.05$).
SO3	G4	To integrate the base yield model and disease-yield loss models into a threescenario forecast engine producing parallel yield predictions (Healthy, Untreated, Treated) with associated revenue projections using market price data from IT21811104 and net treatment benefit calculations.	Scenario engine generates all three forecasts within 3 seconds of receiving input data; net benefit calculation validated against at least two historical disease outbreak seasons.

SO4	G4	To develop a React.js web dashboard that displays real-time three-scenario yield and revenue comparisons, net treatment benefit indicators, and treatment decision recommendations integrated with the group unified platform via REST API in Sinhala and English.	Dashboard passes usability testing with >= 80% task completion rate among 10 Sri Lankan paddy farmer participants; API response time <= 3 seconds.

Table 03: Functional Requirements

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, combining iterative ML model development, experimental validation, and system engineering. DSR is appropriate because the primary output is a working artifact (the yield estimation system) whose quality is evaluated against defined performance criteria. The approach proceeds across five phases, each mapped to a specific objective.

3.1 Research Approach

The research follows the DSR cycle: (1) Problem identification and motivation (literature review and gap analysis. Section 1.2-1.3), (2) Definition of objectives (Section 2), (3) Design and development (Phases 2-4 below), (4) Demonstration (prototype system), and (5) Evaluation (model validation and farmer usability testing). This cycle ensures that the artifact design remains grounded in the identified research gap and that evaluation criteria are pre-defined and measurable.

3.2 Phase 1 - Data Collection and Preprocessing

Three data streams will be assembled. First, real-time IoT data (soil moisture, temperature, humidity, NPK, pH, water level, growth stage) will be received via REST API from IT22178718. Second, historical paddy yield data will be sourced from the Sri Lanka Department of Census and Statistics (DCS) district-level seasonal records (1982–2024, Maha and Yala) and the Hector Kobbekaduwa Agrarian Research Institute market price datasets providing approximately 3,000 historical yield records as the primary training corpus for SO1. Third, IRRI Standard Evaluation System (SES) disease severity-to-yieldloss tables will be digitized for Rice Blast and Brown Spot across five severity grades and five growth stages, producing approximately 50 calibrated reference points as training data for SO2. Data augmentation techniques will expand the effective training dataset to the 400,000+ range required for robust ML model performance, in line with supervisor guidance.

All incoming data will be pre-processed through a validation pipeline including outlier detection (IQR method), missing value imputation (median for sensor data, forward-fill for time-series), and normalization (min-max scaling for ML inputs). Ethical approval for any primary data collection involving farmers will be sought through the SLIIT Research Ethics Committee.

3.3 Phase 2 - Base Yield Estimation Model

A Random Forest (RF) and XGBoost ensemble model will be trained on historical DCS district yield records matched with corresponding IoT sensor and climate feature vectors. Feature engineering will incorporate growth stage (encoded as ordinal: germination, tillering, panicle initiation, heading, grain filling), paddy variety (BG 300, AT 362), and seasonal parameter (Maha/Yala) as primary predictors alongside IoT sensor readings. Hyperparameter tuning will be performed via 5-fold cross-validation on the training set. Model performance will be reported on a held-out test set using RMSE (kg/acre), MAE (kg/acre), and R-squared. The success criterion for SO1 is R-squared >= 0.85.

3.4 Phase 3 - Disease-Yield Loss Regression Models

Separate polynomial regression models will be trained for Rice Blast and Brown Spot using IRRI SES severity grade to yield loss percentage reference data, stratified by growth stage

at time of detection. A severity-by-growth-stage interaction term will be included to capture the finding (Gumma et al. [11]) that growth stage moderates yield loss severity. Models will be fine-tuned using available Sri Lankan district-level disease outbreak records to adjust for local pathotype variations and BG 300/AT 362 susceptibility profiles. Validation will use leave-one-out cross-validation on IRRI SES benchmark data, with success criterion R-squared >= 0.80.

3.5 Phase 4 - Scenario Engine, API, and Dashboard

The three-scenario forecast engine will integrate SO1 and SO2 model outputs to compute parallel yield predictions. Scenario yields are derived as: Healthy = base yield; Untreated = base yield × (1 – disease loss %); Treated = base yield × (1-residual post-treatment loss %). Revenue projections multiply each scenario yield by current market price from IT21811104. Net treatment benefit = (Treated Revenue-Untreated revenue) treatment cost. A Fast API backend will expose forecast outputs via REST API. A React.js dashboard with Chart.js visualizations will present three-scenario comparison charts, revenue projections, net benefit indicators, and treatment decision recommendations in Sinhala and English.

3.6 Tools and Technologies

Category	Tool/ Library	Justification
ML Development	Python 3.11, scikit learn, XGBoost	Industry standard opensource ML libraries with strong documentation and community support
Data Processing	Pandas, NumPy, Matplotlib	Standard data wrangling and visualization stack for ML piplines
Model Training Data	IRRI SES(free), DCS Sri Lanka (Open government data)	Publicly available, authoritative, and directly applicable to Sri Lankan context
Backend API	FastAPI (Python)	High performance async REST framework; wellsuited for ML model serving with minimal overhead
Frontend Dashboard	React.js, Chart.js	Component-based UI framework enabling interactive, responsive three-scenario visualization
Cross-Module Integration	REST APIs (HTTP/HTTPS, JSON)	Standardized inter-module communication with JWT authentication for security

Database	PostgreSQL/ Firebase	Reliable relational storage for yield history; Firebase real-time DB for live dashboard updates
Multilingual UI	i18next (React)	Widely adopted React internationalization library for Sinhala/English support
Deployment	Docker, AWS Free, Tier	Containerized deployment ensuring consistent environment across development and production

3.7 Validation Strategy and Performance Metrics

Model validation will proceed at three levels. At the model level, base yield and diseaseyield loss models will be validated against held-out test sets using quantitative metrics (RMSE, MAE, R-squared, confidence intervals). At the system level, scenario engine outputs will be back-tested against at least two historical Rice Blast and Brown Spot outbreak seasons in Sri Lanka where both disease severity and final yield data are available. At the user level, dashboard usability testing will be conducted with a minimum of 10 Sri Lankan paddy farmers in the target districts of Kurunegala and Anuradhapura using structured task observation sessions to evaluate comprehension and practical utility of the scenario outputs.

3.8 Data Collection and Ethical Considerations

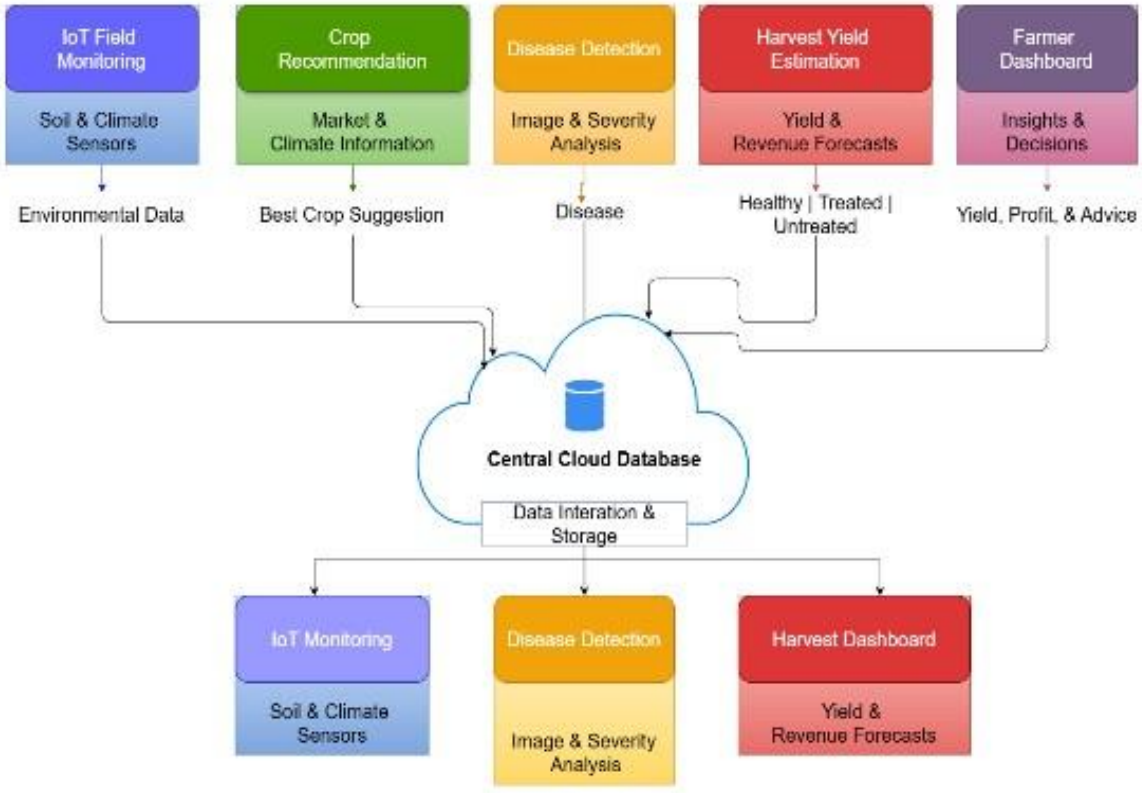
All external datasets (DCS, IRRI SES) are publicly available and accessed through official institutional channels-no data ownership concerns arise. Farmer participation in usability testing will be voluntary with full informed consent obtained before sessions. No personally identifiable information will be stored in the system database. SLIIT Research Ethics Committee approval will be sought prior to any primary data collection involving human participants.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

The proposed system is a four-module Smart Agriculture DSS for Sri Lankan Paddy Cultivation (Project R26-IT-066). Each module addresses a distinct problem domain and communicates with others via REST APIs through a shared cloud layer. The four modules and their relationships are:

- IT22178718 - Context-Aware IoT Field Monitoring: Continuously collects realtime soil and environmental sensor data (moisture, pH, NPK, temperature, humidity, water level). Sends processed sensor readings to this component (IT22266446) as inputs to the base yield estimation model.
- IT21811104 - Intelligent Crop Recommendation: Analyses IoT and climate data against market supply/demand forecasts. Sends current paddy market price per kg to this component for revenue projection calculations within the scenario engine.
- IT22261632 - Disease Detection and Severity Analysis: Detects Rice Blast and Brown Spot from leaf images and environmental sensor data. Estimates disease severity percentage and growth stage at detection. Sends disease type, severity score, and growth stage to this component as inputs to the disease-yield loss regression models.
- IT22266446 - Disease-Aware Harvest Yield Estimation: Receives IoT data, disease severity scores, and market price from the three other modules. Applies the base yield model and disease-yield loss models to generate three-scenario yield and revenue forecasts with net treatment benefit calculation. Delivers outputs via REST API to the unified group dashboard.



4.2 Individual Component Architecture

The Harvest Yield Estimation component consists of four internal processing layers, each directly aligned with a specific objective:

Data Ingestion and Validation: Receives real-time IoT sensor readings from IT22178718 and disease type, severity score, and growth stage from IT22261632 via REST API at configurable polling intervals (default: 15 minutes). Receives current paddy market price from IT21811104 on demand. Applies the preprocessing pipeline (outlier detection, missing value imputation, normalization) before passing validated data to the yield estimation engine.

Yield Estimation Engine: The base yield model (RF/XGBoost ensemble) processes validated IoT sensor data and growth stage to compute baseline expected yield in kg/acre for the current field and variety. The disease-yield loss model (Rice Blast or Brown Spot regression, selected based on IT22261632 output) processes disease severity percentage and growth stage at detection to compute yield loss percentage with confidence interval. Three scenario yields are computed as described in Section 3.5.

Revenue and Decision Calculator: Multiplies each scenario yield by current market price from IT21811104 to generate three revenue projections. Retrieves treatment cost from the DOA pesticide application cost reference table stored in the system database. Computes net treatment benefit and generate a binary decision recommendation (Treat / Do Not Treat) with financial justification string for display.

Dashboard and REST API Layer: A FastAPI backend exposes scenario forecast outputs via authenticated REST API endpoints for consumption by the group unified dashboard. A React.js frontend dashboard component renders three-scenario yield bars, revenue comparison cards, net benefit indicator, and decision recommendation in both Sinhala and English. The dashboard auto-refreshes when new IoT or disease data is received.

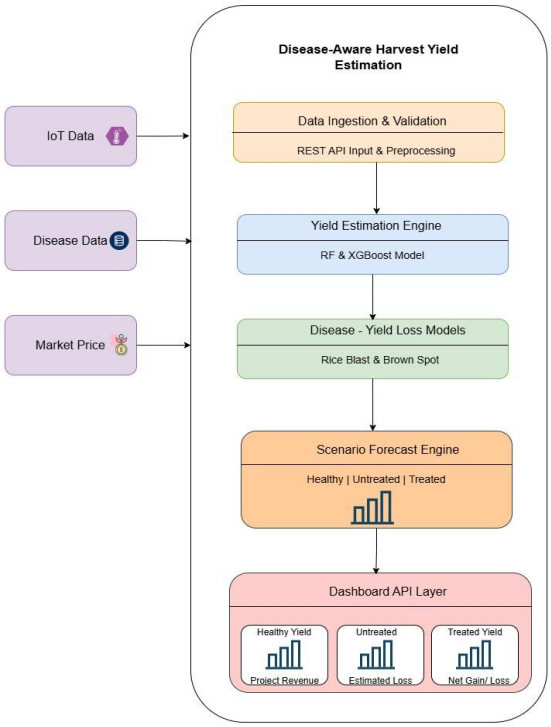


Figure 02: Individual Individual Component Architecture

4.3 Component Interactions and Data Flow

Source Module	Data Set	Used In Layer	Overlap Avoided Because
IT22178718 - IoT	Soil Moisture, temperature, humidity, NPK, Ph, water level, growth stage	Layer 1(ingestion) → Layer 2 (base yield model input)	IT22178718 collects and alerts on sensor thresholds. This component receives those readings and uses them as features in a yield regression model. No duplication of sensing or alert logic.
IT22261632 - Disease	Disease type (Blast / Brown Spot), severity % (0-	Layer 1(ingestion) → Layer 2	IT22261632 detects and classifies diseases. This component receives
	100%), growth stage at detection	(disease-yield loss model input)	severity scores and translates them into yield loss percentages. No duplication of detection or classification logic.
IT21811104 - Market	Current paddy market price per kg	Layer 3 (revenue calculator)	IT21811104 forecasts supply, demand, and market price. This component uses the price output as a multiplier for revenue calculation. No duplication of market forecasting logic.

4.4 Scalability and Security Considerations

The backend API is containerized using Docker to enable horizontal scaling when concurrent user load increases. JWT-based authentication is implemented on all API endpoints to prevent unauthorized cross-module data access. All farmer yield and profile data is stored with AES-256 database-level encryption. The React.js dashboard is designed with responsive layout to support both desktop and mobile form factors, accommodating rural farmers accessing the system via mobile phones on 3G/4G connections.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Users: Smallholder and medium-scale paddy farmers in Sri Lanka (target: 1-5 acres, high-disease-risk districts - Kurunegala, Polonnaruwa, Anuradhapura)
- Secondary Users: Agricultural extension officers using yield trend data for districtlevel advisory
- System Consumers: IT22261632 (provides disease severity inputs), IT22178718 (provides IoT sensor inputs), IT21811104 (provides market price inputs)
- Administrative Users: System administrators managing model retraining pipelines and user account management

5.2 Functional Requirements

ID	Title	Requirement
FR-01	IoT Data Reception	The system shall receive real-time IoT sensor data including soil moisture, temperature, humidity, NPK levels, pH, and water level from IT22178718 via an authenticated REST API endpoint at a configurable polling interval.
FR-02	Disease Severity Input	The system shall receive disease type (Rice Blast / Brown Spot / Healthy), severity score as a percentage (0-100%), and growth stage at time of detection from IT22261632 via an authenticated REST API endpoint.
FR-03	Market Price Input	The system shall receive current paddy market price per kilogram from IT21811104 via REST API for use in scenario revenue projection calculations.
FR-04	Base Yield Estimation	The system shall estimate base paddy harvest yield in kg per acre using the trained RF/XGBoost model with IoT sensor data and growth stage as input features.
FR-05	Disease Yield Loss Calculation	The system shall compute yield loss percentage for Rice Blast and Brown Spot conditions using the respective diseaseyield loss regression model, stratified by growth stage at time of detection.
FR-06	Three Scenario Yield Forecast	The system shall generate three parallel scenario yield forecasts: Scenario A

		(Healthy; no disease), Scenario B (Disease detected and untreated), Scenario C (Disease detected and treated).
FR-07	Revenue Projection	The system shall calculate revenue projections for each scenario by multiplying scenario yield by the current market price from IT21811104.
FR-08	Net Treatment Benefit and Decision	The system shall compute the net financial benefit of treatment as: (Scenario C revenue; Scenario B revenue) minus treatment cost, and generate a treatment decision recommendation (Treat / Do Not Treat) with financial justification.
FR-09	REST API Output	The system shall expose all scenario forecast outputs via authenticated REST API endpoints for consumption by the group unified dashboard.
FR-10	Multilingual Dashboard Display	The dashboard shall display threescenario yield comparison bar charts, revenue comparison cards, net benefit indicator, and treatment recommendation in both Sinhala and English.

Table 03: Functional Requirements

5.3 Non-Functional Requirements

ID	Title	Requirement
NFR-01	Base Model Accuracy	The base yield prediction model shall achieve a minimum R-squared of 0.85 when evaluated on the held-out Sri Lanka DOA district yield validation set.
NFR-02	Disease Model Accuracy	The disease-yield loss regression models (Rice Blast and Brown Spot separately) shall each achieve a minimum Rsquared of 0.80 on IRRI SES benchmark validation data.
NFR-03	Response Time	The three-scenario yield generation pipeline shall produce all outputs within 3 seconds of receiving complete input data from all source modules.
NFR-04	API Security	All REST API endpoints shall implement HTTPS transport and JWT bearer token authentication; unauthenticated requests shall return HTTP 401.

NFR-05	Usability	The dashboard shall achieve a task completion rate of 80% or above in
		structured usability testing with 10 Sri Lankan paddy farmers.
NFR-06	System Availability	The system shall maintain 99% uptime during the active Maha (October–March) and Yala (April–September) cultivation seasons.
NFR-07	Data Security	All farmer yield records and personal data shall be stored with AES-256 encrypted database access; no plaintext personal data shall exist in the system.
NFR-08	Scalability	The system shall handle a minimum of 500 concurrent farmer users without API response time exceeding 3 seconds.

Table 04: Non-Functional Requirements

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary market is Sri Lanka's approximately 1.8 million paddy farming households [2], with an initial focus on medium-scale farmers (1-5 acres) in the high-disease-incidence districts of Kurunegala, Polonnaruwa, and Anuradhapura. Customer persona: male or female farmer, 35-60 years, owns or leases 2-4 acres of paddy, cultivates both Maha and Yala seasons, owns a basic smartphone, has experienced at least one significant diseaserelated yield loss in the past three years. Secondary markets include agricultural insurance providers and rural financial institutions requiring yield loss quantification.

6.2 Value Proposition

This system provides the first quantified, real-time answer to the decision Sri Lankan paddy farmers face every season when a disease is detected: 'Does treating this disease now recover more yield than the treatment costs?' By expressing that answer in specific kg and LKR values across three scenarios, the system converts agronomic intelligence into financial decision support. A capability that no existing Sri Lankan platform provides.

6.3 Revenue Model and Pricing Strategy

- Freemium: Basic healthy yield estimate and seasonal trend dashboard. Free for all registered farmers. Customer acquisition channel.
- Premium Individual (LKR 500-1,000/month): Full three-scenario disease comparison, net treatment benefit calculator, historical yield trend analysis, SMS alerts for critical disease thresholds.
- Cooperative/Institution License (LKR 50,000-150,000/year): Access for agricultural cooperatives or government extension offices serving groups of registered farmers.
- InsurTech API Access (LKR 200,000+/year): Sell yield loss estimation API access to agricultural insurance providers for automated crop damage claims assessment. High-value B2B revenue stream.

6.4 Cost Estimation

Development cost is estimated at LKR 10,000-15,000 one-time (domain and SSL), with operational costs of LKR 3,000-5,000 per month for cloud hosting during the research phase. The majority of development uses open-source tools and shared group infrastructure, minimizing marginal costs.

6.5 Competitive Advantage

No existing Sri Lankan agricultural platform provides disease-scenario harvest yield estimation calibrated for local varieties, diseases, and seasonal patterns. First-mover advantage is reinforced by the integration dependency on the group's IoT and disease detection modules which form a proprietary data supply chain not available to generic global platforms.

6.6 Intellectual Property Considerations

The disease-aware three-scenario yield estimation engine and the Sri Lanka-calibrated disease-yield loss regression models constitute original intellectual property. A provisional patent application will be explored through SLIIT's technology transfer office. Commercial licensing rights are reserved for the development team; an open academic license will be issued for non-commercial research reuse.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost	Justification
Field Data Collection - Farmer Usability Testing (2 field visits, Kurunegala / Anuradhapura)	LKR 12,000	Transport, accommodation, and logistics for structured usability testing sessions with a minimum of 10 smallholder paddy farmers. Required to validate NFR05 task completion rate (≥ 80%) and assess Sinhala-language dashboard comprehension.
IRRI SES Dataset Digitisation and Preprocessing (research assistant — 20 hours)	LKR 15,000	Manual extraction and digitisation of IRRI Standard Evaluation System disease severity-to-yield-loss tables for Rice Blast and Brown Spot across growth stages. Required to build training data for SO2 disease-yield loss regression models.
Domain Registration and SSL Certificate (1 year)	LKR 7,500	Production domain and HTTPS SSL certificate required for live demo deployment of the harvest yield estimation dashboard for evaluation panel review and stakeholder demonstration.
Research and Domain Expert Consultations	LKR 8,000	Structured consultations with DOA agricultural officers and agronomy specialists to validate disease-yield loss model calibration for Sri Lankan paddy varieties BG 300 and AT 362 and local pathotype characteristics.
Printing, Stationery, and Field Documentation (consent forms, logbooks, field survey instruments)	LKR 4,500	Contingency allocation for unforeseen field coordination costs, additional cloud usage beyond planned tier limits, or supplementary data collection requirements identified during development.
Miscellaneous (contingency - communication, field coordination, unexpected expenses)	LKR 5,000	Government open data. Accessible free of charge through official DCS portal.
Cloud Hosting - AWS Free Tier / Firebase (6-month subscription)	LKR 20,000	Scalable cloud infrastructure for ML model inference API, real-time IoT data ingestion, and React.js dashboard hosting. Firebase free tier used initially; paid tier activated if data volume exceeds free limits.
Total	LKR 72,000	

Table 03: Project Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WP	Work Package	Key Tasks	Deliverable
WP1	Data Collection & API Integration (Week 1-4)	Establish mock API contracts with IT22178718,IT21811104,IT22261632. Collect IRRI SES data, DCS yield records, Hector Kobbekaduwa market data. Build preprocessing pipeline.	Validated data pipeline; API integration layer; preprocessed training datasets.
WP2	Base Yield Model (Week 5-9)	Feature engineering; RF and XGBoost training; Hyperparameter tuning via 5fold CV; validation against DCS district yield records	Trained base yield model with R-squared >= 0.85; documented model card
WP3	Disease-Yield Loss Models (Week 1014)	Separate Rice Blast and Brown Spot regression model development; IRRI SES calibration; Sri Lankan variety fine-tuning; LOOCV validation.	Two validated diseaseyield loss models; documented calibration report
WP4	Scenario Engine & API (Weeks 15–19)	Three-scenario forecast engine integration; revenue calculator; net benefit logic; FastAPI REST endpoint development; back-testing.	Working scenario engine API; back-test validation report
WP5	Dashboard & Farmer Testing (Weeks 20–24)	React.js dashboard development; Chart.js visualizations; Sinhala/English i18n; group dashboard integration; 10-farmer usability study.	Production-ready dashboard; usability test report; final integrated system

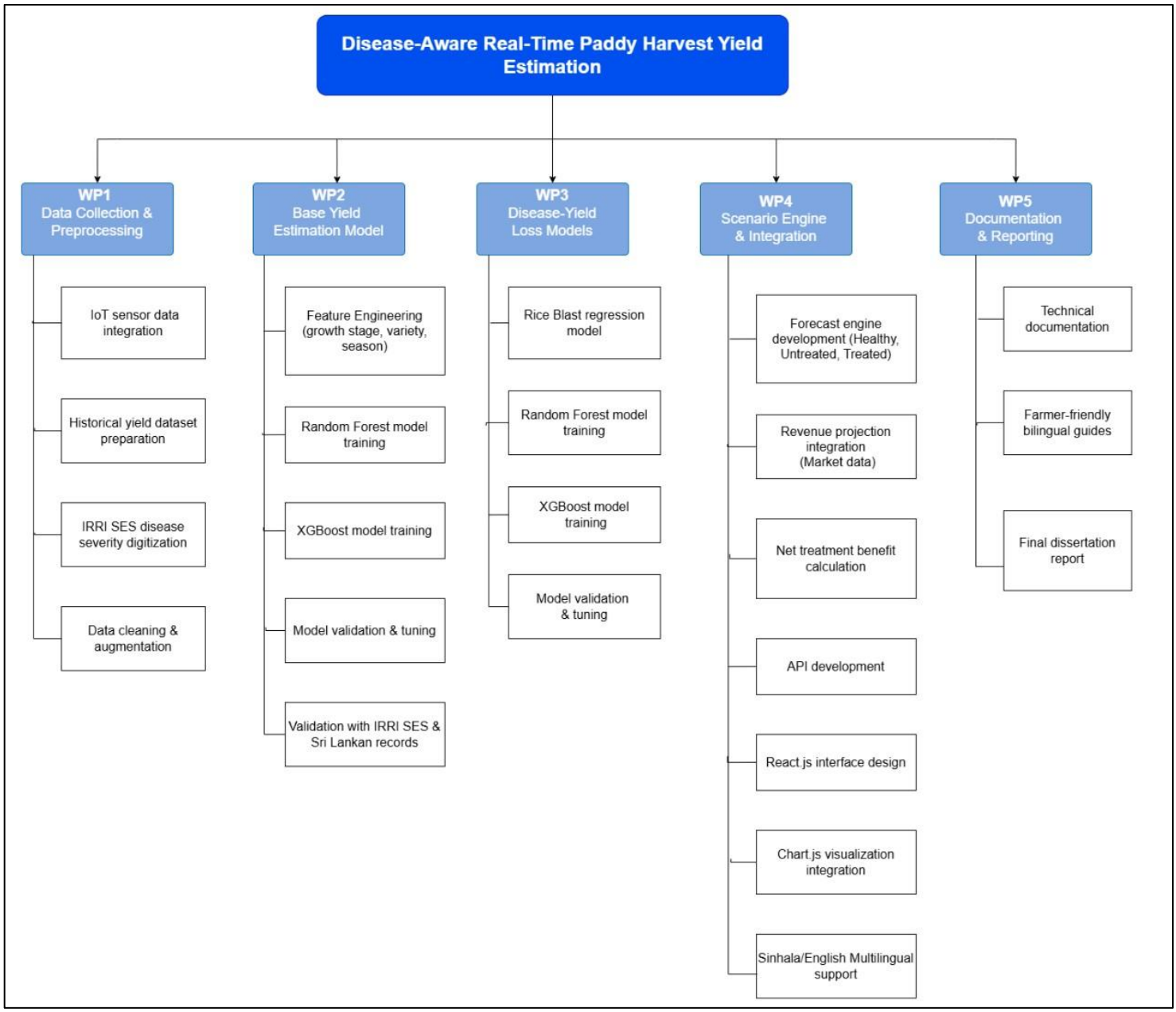


Figure 03: Work Breakdown Structure

9. GANTT CHART

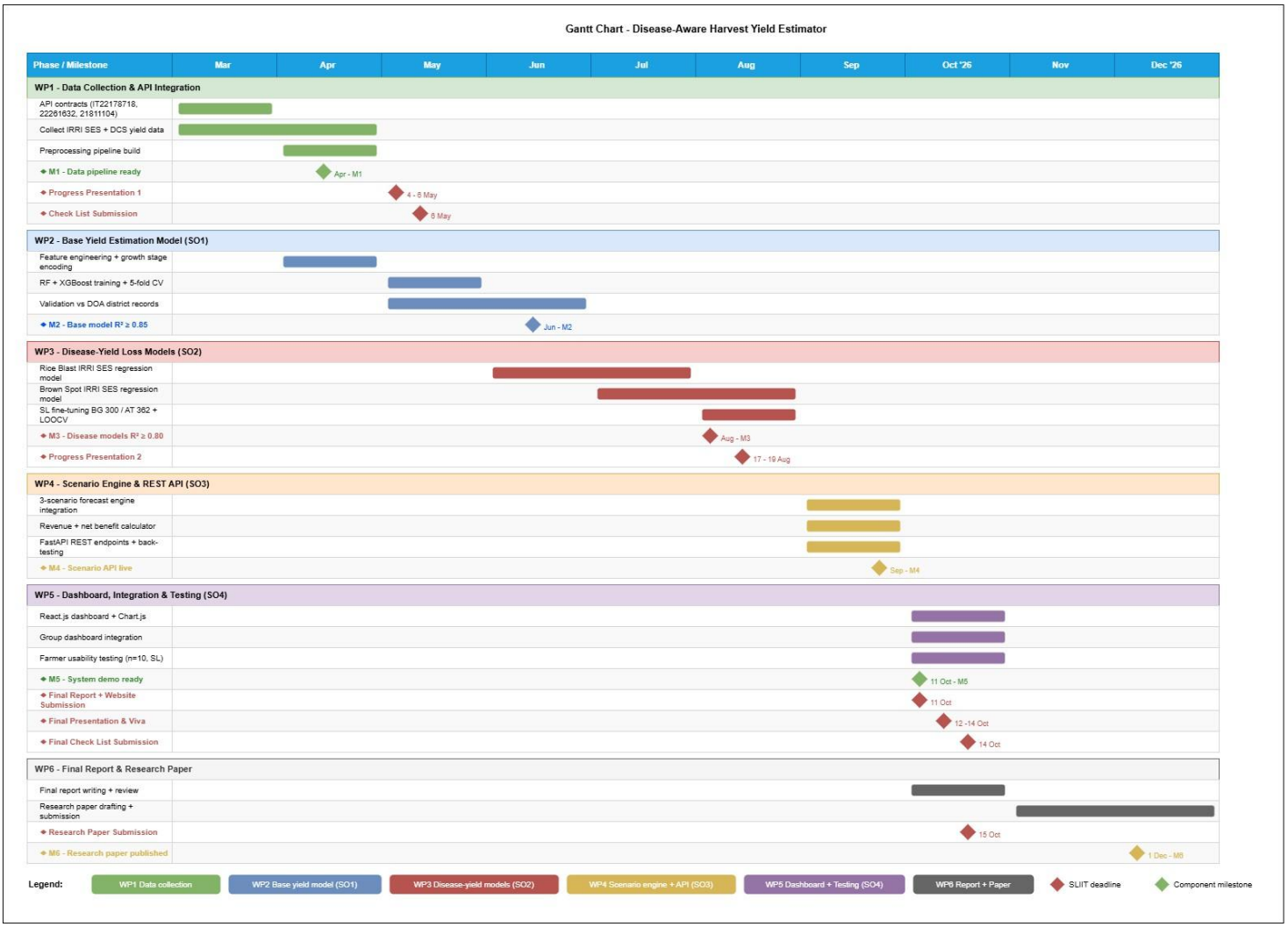


Figure 04:Gantt Chart

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APPENDICES

Appendix A - Rice Blast Yield Loss Reference Table (IRRI SES)

Source: IRRI Standard Evaluation System for Rice, 5th ed., 2013. Used as primary training data for SO2 disease-yield loss model.

Severity Grade	Severity %	Growth Stage at Detection	Estimated Yield Loss %
1	1-5%	Vegetative	2-5%
3	6-25%	Vegetative/Tillering	8-15%
5	26-50%	Tillering/Panicle Initiation	15-25%
7	51-75%	Heading/Flowering	25-40%
9	76-100%	Heading/Flowering (severe)	40-100%

Table 06: Rice Blast Yield Loss Reference

Appendix B - Brown Spot Yield Loss Reference Table (IRRI SES)

Source: IRRI Standard Evaluation System for Rice, 5th ed., 2013. Used as primary training data for SO2 disease-yield loss model.

Severity Grade	Severity %	Growth Stage at Detection	Estimated Yield Loss %
1	1-5%	Vegetative	1-5%
3	6-25%	Tillering	5-15%
5	26-50%	Panicle Initiation	15-25%
7	51-75%	Heading / Flowering	25-35%
9	76-100%	Grain Filling	35-50%

Table 07: Brown Spot Yield Loss Reference

Appendix C - Dataset Sources and Access Details

Dataset	Source	Access Method	Used For
Paddy yield by district/season (1982–2024)	DCS Sri Lanka	Free download: statistics.gov.lk	SO1 base yield model training
IRRI SES diseaseyield loss tables	IRRI	Free PDF: irri.org	SO2 disease-yield loss model training
Paddy market price data	Hector Kobbekaduwa Institute	Request via institution	SO3 revenue projection
Real-time IoT sensor data	IT22178718 (group module)	REST API (internal)	SO1 live model input
Disease severity scores	IT22261632 (group module)	REST API (internal)	SO2 live model input
Current market price	IT21811104 (group module)	REST API (internal)	SO3 revenue calculation

Appendix D — Risk Analysis

Risk	Level	Impact	Mitigation
Disease-yield loss model accuracy below R-squared 0.80	Medium	SO2 success criterion not met	Fine-tune with additional Sri Lankan outbreak data; consult DOA agronomist for expert calibration
IT22261632 API unavailable during development	Medium	SO2, SO3 blocked without live disease severity data	Develop with mock API; define API contract early with IT22261632 member
DCS yield data incomplete for some districts/varieties	Medium	SO1 training data insufficient	Supplement with IRRI Asian yield benchmark data; apply transfer learning
IoT sensor data quality issues from IT22178718	Low	SO1 model receives noisy inputs	Implement preprocessing outlier detection; flag lowquality readings in dashboard
Farmer low digital literacy in usability testing	Low	NFR05 task completion rate below 80%	Conduct Sinhala-language guided walkthroughs before independent tasks

Appendix E — Ethical Considerations

- All external datasets (DCS, IRRI SES) are publicly available through official channels. No personal data is involved in obtaining these datasets.
- Farmer participation in usability testing is entirely voluntary. Written informed consent will be obtained before any testing session.
- No personally identifiable information (name, NIC, location) will be stored in the production system database. Farmer profiles use anonymized IDs.
- SLIIT Research Ethics Committee approval will be obtained prior to conducting any primary data collection or farmer usability testing involving human participants.
- AI writing assistance tools were used to support drafting and structuring of this report. All research direction, component design decisions, novelty justification, and analytical interpretation are the original work of the student.